Exact-Content Progression Normalization and Resonance:

Fixed-Determinant Recovery, Conductor Sectors, and Multiplier-Mass Accounting

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Abstract

TPC-93 resolves the exact-content condition

$$b = c\kappa \mid \sigma U(t)$$

by writing $t = \tau + Bz$, where $B = b/(b, \sigma)$. This paper audits the two structures created by that substitution: the affine determinant and the additive phase.

For an individually primitive pair

$$D(t) = d + \sigma t, \qquad U(t) = u + at, \qquad \sigma u - ad = h_0 \neq 0,$$

a soluble content progression gives

$$U(\tau + Bz) = B\,V(z).$$

The unreduced pair $(D(\tau + Bz), U(\tau + Bz))$ has determinant Bh_0, whereas the semi-reduced pair $(D(\tau + Bz), V(z))$ has determinant h_0. In the general abstract setting the first form can acquire the exact content

$$H = \gcd(D(\tau), \sigma B) = \gcd(B, h_0).$$

After extracting it, the primitive pair has determinant h_0/H, and

$$\mu(D(\tau + Bz))\mu(U(\tau + Bz)) = \mu(H)\mu(B)\mu(\widetilde{D}(z))\mu(V(z))\,\mathbf{1}_{(H,\widetilde{D}(z))=1}\,\mathbf{1}_{(B,V(z))=1}.$$

Thus squareful B vanishes identically, and the apparent growing determinant Bh_0 is a representation effect, not by itself a growing-determinant obstruction. On the literal canonical TPC-32 support, b is a common target divisor and $(b, h_0) = 1$; hence $H = 1$ and the reduced primitive determinant is exactly the prescribed h_0.

The determinant character becomes

$$\zeta_\xi \exp\!\left(-2\pi i\,\frac{\varepsilon_\xi \widetilde{r}_\xi \ell_\xi v_\xi \sigma_\xi B_\xi}{c_\xi q_X} z\right).$$

Here $\varepsilon_\xi \in \{+1, -1\}$ records the source polarization, and $\widetilde{r}_\xi$ is the unique least-absolute signed lift of the low-window class $r_\xi \bmod q_X$. Its exact conductor is

$$q_\xi^* = \frac{c_\xi q_X}{\gcd(c_\xi q_X, \widetilde{r}_\xi \ell_\xi v_\xi \sigma_\xi B_\xi)}.$$

For prime q_X, $c_\xi < q_X$, and nonzero low-window r_ξ, there is a conductor gap: if q_X divides the phase slope then $q_\xi^* \le c_\xi$, while otherwise $q_\xi^* \ge q_X$. Exact constancy, low conductor, archimedean resonance, and generic oscillation are therefore different predicates.

The literal common-target provenance is stronger:

$$b_\xi \mid M_\xi(z) - n_\xi, \qquad b_\xi \mid \ell_\xi v_\xi \sigma_\xi B_\xi.$$

Hence c_ξ divides the phase slope, and the conductor is exactly

$$q_\xi^* \in \{1, q_X\}.$$

It equals 1 precisely on the constant branch $q_X \mid \ell_\xi v_\xi \sigma_\xi B_\xi$; otherwise it equals q_X. Thus no nonconstant low-conductor sector survives in the literal packet.

We give a mutually exclusive and exhaustive abstract partition into null, short, long constant, long low-conductor resonant/generic, and long high-conductor resonant/generic sectors, and prove its four-sector literal collapse. Each exceptional return is charged by

$$\sum_{r \text{ in the sector}} |\mathfrak{m}_{K,X}(r)|,$$

not by the number of frequencies. All determinant, conductor, and partition results are L0; by the complete lossless provenance theorem established in TPC-93, their identification with that literal packet is L1. No L2 cancellation, fixed-shift prime-pair estimate, parity breach, or twin-prime conclusion is proved.

Keywords: exact content; affine progression; fixed determinant; Möbius factorization; additive conductor; resonance; short fiber; multiplier mass.

Literal-frame convention. Every physical statement retains one prescribed $h_0 \neq 0$, the actual content variables c, κ, both polarizations, every native interval and mask, the signed reconstruction multiplier, and one global post-aggregation normalization. The orientation ε_θ and the unique least-absolute signed lift of each low-window class are never suppressed. No statement is specialized to $h_0 = 2$.

1 Introduction

The low-window export of TPC-93 writes the distinguished zero-frequency functional as one coherent sum over native affine keys, exact contents, inclusion–exclusion variables, and Fourier frequencies [5]. One inner term has the shape

$$\begin{aligned}\sum_{t \in I_\theta} &\mu(D_\theta(t))\mu(U_\theta(t)) W_{\theta,X}(t) \mathbf{1}_{c\kappa \mid \sigma_\theta U_\theta(t)} \\ &\times e_{cq_X}\left(-\varepsilon_\theta \widetilde{r}(M_\theta(t) - n_\theta)\right),\end{aligned} \tag{1}$$

where

$$D_\theta(t) = d_\theta + \sigma_\theta t, \qquad U_\theta(t) = u_\theta + a_\theta t,$$
$$\sigma_\theta u_\theta - a_\theta d_\theta = h_0, \qquad M_\theta(t) = \ell_\theta v_\theta D_\theta(t).$$

Here $\varepsilon_\theta = +1$ on the left polarization and $\varepsilon_\theta = -1$ on the right polarization. For $r \in \mathscr{W}_X(R) \subset \mathbb{Z}/q_X\mathbb{Z}$, $\widetilde{r}$ denotes its unique signed lift with $-q_X/2 < \widetilde{r} < q_X/2$. This lift, rather than an unspecified integer representative of r, is used in every character modulo cq_X. All omitted factors in (1) are fixed by the outer key; no fiber is renormalized separately.

Put $b = c\kappa$, $g = (b, \sigma_\theta)$, and $B = b/g$. The content condition reduces to $B \mid U_\theta(t)$. On a soluble branch it selects one class $t = \tau + Bz$. Two questions must then be kept separate.

Which determinant is relevant? If one merely substitutes $t = \tau + Bz$, both slopes gain a factor B, and the raw determinant is Bh_0. But the congruence itself forces $U(\tau + Bz) = BV(z)$. Extracting that factor returns a determinant-h_0 pair, together with an outer sign and a coprimality mask. A further fixed content can occur in the first form outside the canonical TPC-32 support. The exact answer is the finite palette h_0/H, $H = (B, h_0)$, rather than an uncontrolled growing determinant.

Which phase is relevant? Content-factor extraction does not divide the determinant character. The phase increment remains

$$\alpha_\xi = \frac{\varepsilon_\xi \widetilde{r}_\xi \ell_\xi v_\xi \sigma_\xi B_\xi}{c_\xi q_X} \pmod 1.$$

Its rational conductor, its distance to the nearest integer, and the resolved fiber length encode three distinct mechanisms: periodicity, near constancy, and shortness.

1.1 Main results

The first theorem gives the full two-stage normalization:

$$\begin{array}{ccl} (D(\tau+Bz), U(\tau+Bz)) & : & \det = Bh_0, \\ (D(\tau+Bz), V(z)) & : & \det = h_0, \\ (\widetilde{D}(z), V(z)) & : & \det = h_0/H, \quad H = (B, h_0). \end{array}$$

It also gives the exact Möbius masks at both extractions. The literal common-target provenance of TPC-32 forces $H = 1$ [1].

The second theorem computes the exact additive conductor. Because the TPC-86 modulus is prime and the content c is smaller than that modulus, every nonconstant phase has either conductor at most c or conductor at least q_X; there is no middle range [2]. On the literal common-target branch one has the stronger divisibility $c \mid \Omega$, so the two possibilities collapse to conductor 1 and conductor q_X.

The third theorem gives a disjoint and exhaustive sector partition. Its natural specialization uses a short-fiber threshold at least as large as the content cutoff. In that regime a long nonconstant low-conductor fiber cannot also be resonant. In the literal packet there is no nonconstant low-conductor fiber at all.

Finally, we formulate the exact multiplier-mass ledger. A set of exceptional frequencies is inexpensive only when its reconstruction-multiplier mass is inexpensive after the literal outer coefficients and fiber masses are restored. Cardinality by itself is not the relevant quantity.

1.2 Proof levels

Affine substitution, content extraction, conductor reduction, fiber counting, and finite set partitions are L0. Verifying that b is exactly a canonical common-target divisor and restoring all TPC-93 provenance is an L1 attachment here: it is the direct application of the complete lossless source-to-decorated theorem and its explicit inverse ledger established in TPC-93. Any growing-X saving for a returned sector is L2, and remains open here.

1.3 Organization

Section 2 proves the general normalization theorem. The literal canonical specialization is section 3. Exact phase conductors are computed in section 4; the sector partition appears in section 5. Multiplier-mass accounting and the audited claim boundary occupy sections 6 and 7.

2 Two-stage normalization on a content progression

We first work in an abstract affine setting. This separates the algebra that is always true from the stronger coprimality supplied by the literal TPC-32 provenance.

2.1 One progression and the forced factor

Let

$$D(t) = d + \sigma t, \qquad U(t) = u + at, \qquad \sigma u - ad = h_0 \neq 0. \tag{2}$$

Assume that the two forms are individually primitive and that

$$(a, \sigma) = 1.$$

For $b \geq 1$, put

$$g = (b, \sigma), \qquad B = b/g. \tag{3}$$

Then

$$b \mid \sigma U(t) \iff B \mid U(t).$$

Because $(u, a) = 1$, the last congruence has no solution if $(a, B) > 1$, and has exactly one residue class modulo B if $(a, B) = 1$. We henceforth work on the soluble branch and choose τ with

$$U(\tau) \equiv 0 \pmod{B}.$$

Set

$$d_B = D(\tau), \qquad v_B = \frac{U(\tau)}{B}, \tag{4}$$

$$D_B(z) = D(\tau + Bz) = d_B + \sigma Bz, \qquad V_B(z) = \frac{U(\tau + Bz)}{B} = v_B + az. \tag{5}$$

The raw second value is $U(\tau + Bz) = BV_B(z)$.

Proposition 2.1 (Raw and semi-reduced determinants)**.** *The unreduced pair*

$$D_B(z), \qquad U(\tau + Bz)$$

has determinant Bh_0. The semi-reduced pair

$$D_B(z), \qquad V_B(z)$$

has determinant h_0:

$$(\sigma B)v_B - ad_B = h_0. \tag{6}$$

Moreover,

$$(v_B, a) = 1, \qquad (a, \sigma B) = 1.$$

Proof. The raw origins are d_B, Bv_B and the raw slopes are $\sigma B, aB$, so the determinant is

$$(\sigma B)(Bv_B) - (aB)d_B = B\{\sigma U(\tau) - aD(\tau)\} = Bh_0.$$

Dividing the second form by B gives (6). Since $(a, B) = 1$,

$$(v_B, a) = (Bv_B, a) = (U(\tau), a) = (u, a) = 1.$$

The last assertion follows from $(a, \sigma) = (a, B) = 1$. □

2.2 The robustness content in the first form

The semi-reduced first form need not be primitive in an arbitrary noncanonical packet.

Theorem 2.2 (Exact robustness content and determinant palette)**.** *Let*

$$H = \gcd(d_B, \sigma B). \tag{7}$$

Then

$$\boxed{H = \gcd(B, h_0).} \tag{8}$$

Writing

$$\widetilde{D}_B(z) = \frac{D_B(z)}{H} = \frac{d_B}{H} + \frac{\sigma B}{H} z,$$

the pair $(\widetilde{D}_B, V_B)$ is individually primitive and has determinant

$$\boxed{\det(\widetilde{D}_B, V_B) = h_0/H.} \tag{9}$$

In particular, as B varies, the primitive determinant belongs to the finite set

$$\{h_0/H : H \mid |h_0|\}.$$

Proof. Translation does not change the content of the first native form, so

$$(d_B, \sigma) = (d, \sigma) = 1.$$

Consequently

$$(d_B, \sigma B) = (d_B, B).$$

The identity (6) gives

$$a d_B = \sigma B v_B - h_0.$$

Since $(a, B) = 1$,

$$(d_B, B) = (a d_B, B) = (h_0, B),$$

which proves (8). Dividing the first row of (6) by H proves (9). The definition of H makes $\widetilde{D}_B$ primitive, while proposition 2.1 makes V_B primitive. □

Remark 2.3 (What is a representation artifact). The determinant Bh_0 belongs to the unreduced pair in which the forced factor B is still inside the second value. It is not the determinant of the two Möbius-variable forms after exact content extraction. The only remaining determinant change is the fixed content $H \mid h_0$, and even that disappears on the literal canonical branch proved in section 3.

2.3 Exact Möbius extraction

Lemma 2.4 (Universal Möbius content identity)**.** *For positive integers m, n,*

$$\mu(mn) = \mu(m)\mu(n)\mathbf{1}_{(m,n)=1}. \tag{10}$$

The formula includes the squareful case: if m is not squarefree, both sides vanish.

Proof. If $(m, n) = 1$, this is multiplicativity. If a prime divides both, then mn is divisible by its square and both sides of (10) vanish because of the indicator. □

Theorem 2.5 (Two-stage Möbius normalization)**.** *On positive values,*

$$\begin{aligned} &\mu(D(\tau + Bz))\mu(U(\tau + Bz)) \\ &\quad = \mu(H)\mu(B)\mu(\widetilde{D}_B(z))\mu(V_B(z))\mathbf{1}_{(H,\widetilde{D}_B(z))=1}\mathbf{1}_{(B,V_B(z))=1}. \end{aligned} \tag{11}$$

Hence:

(i) if B is squareful, the original product vanishes for every z;

(ii) if B is squarefree, then $H \mid B$ is squarefree and the surviving coefficient contains the literal signs $\mu(B)\mu(H)$ and both displayed coprimality masks;

(iii) if $H = 1$, the primitive Möbius pair has determinant exactly h_0.

Proof. Use

$$D(\tau + Bz) = H\widetilde{D}_B(z), \qquad U(\tau + Bz) = BV_B(z)$$

and apply lemma 2.4 twice. The conclusions follow from $H = (B, h_0) \mid B$. □

Remark 2.6 (No mask may be silently deleted). Even when B and H are squarefree, the two coprimality indicators in (11) are genuine periodic arithmetic masks. Absorbing their coefficients into the outer key is harmless; replacing either mask by 1 changes the literal sum.

3 The literal canonical branch

The robustness theorem deliberately did not assume that b came from the TPC-32 canonical common-target projector. We now restore that provenance exactly by importing the established TPC-93 Theorem 3.3, Corollary 3.4, and Theorem 3.5 (Complete Decorated Affine Export), whose explicit inverse and multiplicity reconstruction furnish the required lossless source-to-decorated ledger.

For a left-polarized TPC-93 key,

$$\sigma_\theta U_\theta(t) = M_\theta(t) j_\theta + h_0, \qquad P_\theta = n_\theta j_\theta + h_0.$$

An active content-inversion key satisfies

$$b = c\kappa \mid \sigma_\theta U_\theta(t), \qquad b \mid P_\theta.$$

Thus b is literally a common divisor of the two original targets, not merely a divisor of a reduced surrogate. For a right-polarized key the two row labels are exchanged. The same common-target statements hold, while the determinant-character orientation is recorded separately by $\varepsilon_\theta = -1$.

Theorem 3.1 (Literal removal of the robustness content)**.** *On the TPC-32 primitive target support, import the established TPC-93 Theorem 3.3, Corollary 3.4, and Theorem 3.5, including their lossless inverse provenance and multiplicity reconstruction [5]. Then every active content-inversion key in either polarization satisfies*

$$(b, h_0) = 1. \tag{12}$$

Consequently

$$(B, h_0) = 1, \qquad H = 1.$$

The two affine Möbius arguments after extracting B are individually primitive and have the same prescribed determinant h_0:

$$D_\theta(\tau + Bz), \qquad V_{\theta,b}(z) = \frac{U_\theta(\tau + Bz)}{B}.$$

Proof. The TPC-32 common-target-divisor lemma states that every common target divisor w is coprime to the physical product containing the fixed shift h_0 [1]. Apply it with

$$w = b, \quad N_m = \sigma_\theta U_\theta(t), \quad N_n = P_\theta.$$

This proves (12). Since $B \mid b$, $(B, h_0) = 1$. Now apply theorem 2.2 to obtain $H = 1$ and the primitive determinant h_0. □

Corollary 3.2 (Literal normalized Möbius pair)**.** *On the literal canonical branch,*

$$\begin{aligned} &\mu(D_\theta(\tau + Bz))\mu(U_\theta(\tau + Bz)) \\ &\qquad = \mu(B)\mu(D_\theta(\tau + Bz))\mu(V_{\theta,b}(z))\mathbf{1}_{(B, V_{\theta,b}(z))=1}. \end{aligned} \tag{13}$$

Every squareful B branch is identically zero. Every surviving branch is a primitive determinant-h_0 TPC-90 pair with one extra coprimality mask and the extracted sign $\mu(B)$ [3].

Proof. Set $H = 1$ in (11). □

Proposition 3.3 (Literal content divisibility)**.** *Under the imported lossless-provenance theorem above, for every active canonical content key in either polarization and every $z \in \mathbb{Z}$,*

$$b \mid M_\theta(\tau + Bz) - n_\theta. \tag{14}$$

Consequently

$$b \mid \Omega_{\theta,b} = \ell_\theta v_\theta \sigma_\theta B, \qquad c \mid \Omega_{\theta,b}. \tag{15}$$

Proof. The defining progression gives

$$b \mid M_\theta(\tau + Bz)j_\theta + h_0, \qquad b \mid n_\theta j_\theta + h_0$$

for every integer z. Subtraction gives

$$b \mid \{M_\theta(\tau + Bz) - n_\theta\}j_\theta.$$

The TPC-32 common-target-divisor lemma gives $(b, j_\theta) = 1$, so j_θ may be cancelled, proving (14). Subtract that divisibility at $z+1$ and z. The difference is $\Omega_{\theta,b}$, proving (15). Finally $c \mid b$. The right polarization is identical after exchanging the two row labels; the sign $\varepsilon_\theta = -1$ affects the oriented phase but not any displayed divisibility. □

Remark 3.4 (Why the abstract palette is still needed). The conclusion $H = 1$ depends on three exact provenance facts: σU and P are the original targets, b divides both, and the TPC-32 primitive support has not been enlarged. If any of these facts is lost under a future completion or surrogate, one must return to theorem 2.2; the correct primitive determinant is then h_0/H, not a silently relabeled h_0.

4 The phase after content resolution

The content-factor normalization changes the Möbius arguments but does not divide the determinant character. This is the second provenance boundary.

4.1 Signed frequency lift and polarization

The Fourier window $\mathscr{W}_X(R)$ consists of residue classes modulo the odd prime q_X. For every $r \in \mathscr{W}_X(R)$, let

$$\widetilde{r} \in \mathbb{Z}, \qquad -\frac{q_X}{2} < \widetilde{r} < \frac{q_X}{2}, \qquad \widetilde{r} \equiv r \pmod{q_X}, \tag{16}$$

be the unique signed lift. Thus

$$0 < |\widetilde{r}| = d_{q_X}(r, 0) < q_X/2, \qquad q_X \nmid \widetilde{r}.$$

We retain the source orientation $\varepsilon_\theta \in \{+1, -1\}$. The multiplier $\mathfrak{m}_{K,X}(r)$ remains a function of the residue class; only its lift into a character modulo cq_X uses $\widetilde{r}$. Without (16), a general character modulo cq_X need not be invariant under replacing an integer representative by that representative plus q_X.

4.2 Exact phase increment

Write

$$M_\theta(t) = \ell_\theta v_\theta D_\theta(t).$$

After $t = \tau + Bz$,

$$M_\theta(\tau + Bz) = M_\theta(\tau) + \ell_\theta v_\theta \sigma_\theta Bz.$$

For a resolved key

$$\xi = (\theta, c, \kappa, r), \qquad b = c\kappa,$$

put

$$\Omega_\xi = \ell_\theta v_\theta \sigma_\theta B_{\theta,b}, \tag{17}$$

$$\alpha_\xi = \frac{\varepsilon_\theta \widetilde{r}\Omega_\xi}{cq_X} \pmod 1, \tag{18}$$

$$\zeta_\xi = e_{cq_X}(-\varepsilon_\theta \widetilde{r}(M_\theta(\tau) - n_\theta)). \tag{19}$$

Then

$$e_{cq_X}(-\varepsilon_\theta \widetilde{r}(M_\theta(\tau + Bz) - n_\theta)) = \zeta_\xi e(-\alpha_\xi z), \tag{20}$$

where $e(x) = e^{2\pi i x}$.

Remark 4.1 (Determinant and phase slope are different data)*.* Extracting B from U, or H from D, changes the affine pair used in the Möbius correlation. It does not change Ω_ξ in (17), because the additive character was attached to the original moving row. The arithmetic determinant h_0/H and the additive phase conductor must therefore be recorded separately.

4.3 Exact rational conductor

Definition 4.2 (Resolved additive conductor)**.** The additive conductor of ξ is

$$q_\xi^* = \operatorname{cond}(\alpha_\xi) := \frac{cq_X}{\gcd(cq_X, \widetilde{r}\Omega_\xi)}. \tag{21}$$

It is the least positive period of $z \mapsto e(-\alpha_\xi z)$.

Proposition 4.3 (Constant phase)**.** *The phase in* (20) *is constant in* z *if and only if*

$$q_\xi^* = 1 \quad \Longleftrightarrow \quad cq_X \mid \widetilde{r}\Omega_\xi. \tag{22}$$

This is an exact arithmetic condition, not the inequality defining near resonance.

Proof. The character is constant precisely when $\alpha_\xi = 0$ in $\mathbb{R}/\mathbb{Z}$, which is equivalent to (22). □

Theorem 4.4 (Prime-modulus conductor gap)**.** *Suppose* q_X *is prime,*

$$1 \le c \le C_X < q_X, \qquad 0 < |\widetilde{r}| < q_X/2.$$

Then $q_X \nmid \widetilde{r}$ *and:*

(i) if $q_X \mid \Omega_\xi$*, writing* $\Omega_\xi = q_X\Omega'_\xi$*, one has*

$$q_\xi^* = \frac{c}{(c, \widetilde{r}\Omega'_\xi)} \le c \le C_X; \tag{23}$$

(ii) if $q_X \nmid \Omega_\xi$*, then*

$$q_\xi^* = q_X \frac{c}{(c, \widetilde{r}\Omega_\xi)} \ge q_X. \tag{24}$$

Thus no conductor lies strictly between C_X *and* q_X*.*

Proof. By (16), $q_X \nmid \widetilde{r}$. Also $(c, q_X) = 1$.

If $q_X \mid \Omega_\xi$, cancel q_X from numerator and denominator in (21) to obtain (23). If $q_X \nmid \Omega_\xi$, then $q_X \nmid \widetilde{r}\Omega_\xi$, so

$$(cq_X, \widetilde{r}\Omega_\xi) = (c, \widetilde{r}\Omega_\xi).$$

This gives (24). □

Corollary 4.5 (Explicit constant branch)**.** *Under the hypotheses of theorem 4.4, constancy is possible only when* $q_X \mid \Omega_\xi$*. Writing* $\Omega_\xi = q_X\Omega'_\xi$*, it occurs precisely when*

$$c \mid \widetilde{r}\Omega'_\xi. \tag{25}$$

Proof. Combine proposition 4.3 with (23). □

Theorem 4.6 (Binary conductor on the literal packet)**.** *Use the literal common-target provenance proved in proposition 3.3. Put*

$$\omega_\xi = \Omega_\xi/c \in \mathbb{Z}. \tag{26}$$

Under the prime-modulus hypotheses of theorem 4.4,

$$\boxed{q_\xi^* = \frac{q_X}{(q_X, \widetilde{r}\omega_\xi)} = \begin{cases} 1, & q_X \mid \omega_\xi, \\ q_X, & q_X \nmid \omega_\xi. \end{cases}} \tag{27}$$

Because $(c, q_X) = 1$*, the first case is equivalent to* $q_X \mid \Omega_\xi$*. Thus:*

(i) $q_X \mid \Omega_\xi$ *makes the phase constant for* every *low-window frequency* r*;*

(ii) $q_X \nmid \Omega_\xi$ *gives exact conductor* q_X *for* every *low-window frequency; and*

(iii) no literal nonconstant phase has conductor between 2 *and* $q_X - 1$*.*

Proof. By (15), $c \mid \Omega_\xi$. Cancel c in (21) to get the first equality in (27). The modulus q_X is prime and $q_X \nmid \widetilde{r}$, so the gcd in the denominator is either q_X or 1. Finally $(c, q_X) = 1$ makes $q_X \mid \omega_\xi$ equivalent to $q_X \mid \Omega_\xi$. □

Remark 4.7 (Literal representative invariance). The signed lift is essential for the general abstract phase. On the literal packet, however,

$$c \mid M_\theta(\tau + Bz) - n_\theta, \qquad c \mid \Omega_\xi.$$

Replacing $\widetilde{r}$ by $\widetilde{r} + kq_X$ therefore multiplies the full character by

$$e_c(-k\varepsilon_\theta(M_\theta(\tau + Bz) - n_\theta)) = 1.$$

Thus the literal phase and its binary conductor are genuinely functions of $r \bmod q_X$; the convention (16) additionally makes every abstract statement unambiguous.

Corollary 4.8 (Literal resonance test on the nonconstant branch)**.** *If* $q_X \nmid \Omega_\xi$*, then*

$$\mathcal{R}(\xi) \iff N_\xi \, d_{q_X}(\widetilde{r}\omega_\xi, 0) \le q_X. \tag{28}$$

In particular, a nonconstant resonant fiber has $N_\xi \le q_X$*.*

Proof. Use

$$\|\alpha_\xi\|_{\mathbb{R}/\mathbb{Z}} = \frac{d_{q_X}(\widetilde{r}\omega_\xi, 0)}{q_X}$$

in (29). The last assertion also follows from lemma 4.10 and $q_\xi^* = q_X$. □

Remark 4.9 (Where low conductor can come from). Since

$$\Omega_\xi = \ell_\theta v_\theta \sigma_\theta B_{\theta,b},$$

the condition $q_X \mid \Omega_\xi$ can come either from a native slope factor or from the content step $B_{\theta,b}$. If $q_X \nmid \ell_\theta v_\theta \sigma_\theta$, then it forces $q_X \mid B_{\theta,b}$, which also shortens the resolved fiber by at least a factor q_X. No such size conclusion is valid when q_X already divides a native slope factor.

4.4 Archimedean resonance

Let N_ξ be the cardinality of the resolved z-interval. Define

$$\mathcal{R}(\xi) \iff N_\xi \|\alpha_\xi\|_{\mathbb{R}/\mathbb{Z}} \le 1. \tag{29}$$

Lemma 4.10 (Nonconstant resonance forces a conductor-length bound)**.** *If* $q_\xi^* > 1$ *and* $\mathcal{R}(\xi)$*, then*

$$N_\xi \le q_\xi^*. \tag{30}$$

Proof. In lowest terms $\alpha_\xi = p/q_\xi^*$ with $(p, q_\xi^*) = 1$ and $p \not\equiv 0 \pmod{q_\xi^*}$. Therefore

$$\|\alpha_\xi\|_{\mathbb{R}/\mathbb{Z}} \ge 1/q_\xi^*.$$

Insert this in (29). □

Remark 4.11 (No automatic oscillatory saving). On a constant or resonant fiber the additive phase supplies no uniform cancellation. On a low-conductor fiber it is periodic, but the Möbius signs, coprimality masks, and smooth weights need not be constant on phase periods. Even on a high-conductor generic fiber, partial sums of the full weighted Möbius product still require a new arithmetic estimate. The conductor calculation alone is L0, not L2.

5 Fiber length and an exhaustive sector partition

5.1 Exact resolved length

Let the native interval be

$$I_\theta = [L_\theta, U_\theta] \cap \mathbb{Z}, \qquad N_\theta^{(0)} = |I_\theta|.$$

For a soluble branch, the resolved interval is

$$I_{\theta,b}^* = \{z \in \mathbb{Z} : \tau_{\theta,b} + B_{\theta,b} z \in I_\theta\}.$$

Proposition 5.1 (Progression thinning)**.** *The resolved cardinality is*

$$N_{\theta,b} = \max\left\{0, \left\lfloor \frac{U_\theta - \tau_{\theta,b}}{B_{\theta,b}} \right\rfloor - \left\lceil \frac{L_\theta - \tau_{\theta,b}}{B_{\theta,b}} \right\rceil + 1\right\}, \tag{31}$$

and

$$N_{\theta,b} \le \left\lceil \frac{N_\theta^{(0)}}{B_{\theta,b}} \right\rceil \le \frac{N_\theta^{(0)}}{B_{\theta,b}} + 1. \tag{32}$$

In particular:

(i) if $B_{\theta,b} \ge N_\theta^{(0)}$, the fiber has at most one point;

(ii) for any integer $Y \ge 1$, if $B_{\theta,b} \ge N_\theta^{(0)}/Y$, then $N_{\theta,b} \le Y$;

(iii) if $q_X \nmid \ell_\theta v_\theta \sigma_\theta$, $q_X \mid \Omega_\xi$, and $N_\theta^{(0)} \le q_X$, then the fiber has at most one point.

Proof. The first formula is obtained by solving the two endpoint inequalities for z. An interval of $N_\theta^{(0)}$ consecutive integers contains at most $\lceil N_\theta^{(0)}/B \rceil$ members of one class modulo B, which proves (32). Parts (i) and (ii) follow immediately. In part (iii), the divisibility of $\Omega_\xi = \ell_\theta v_\theta \sigma_\theta B$ forces $q_X \mid B$; now use part (i). □

Remark 5.2 (Short is a physical return, not a deletion)*.* The estimate $N_{\theta,b} \le Y$ does not make the corresponding term zero. It identifies a short-fiber return whose complete outer coefficient, masks, and reconstruction-multiplier mass must still be bounded.

5.2 General disjoint partition

Fix an integer short threshold $Y_X \ge 1$ and a conductor threshold $Q_X^{\mathrm{cond}} \ge 2$. Let Ξ_X be the finite set of all resolved (θ, c, κ, r) keys before estimating any sector. Put in $\mathfrak{N}$ every insoluble key, empty interval, or squareful B branch, as well as every key with $\mu(\kappa) = 0$. Such terms are exactly zero.

For every remaining key define:

$$\begin{aligned}
\mathfrak{S} &= \{\xi : N_\xi \le Y_X\},\\
\mathfrak{C} &= \{\xi : N_\xi > Y_X,\ q_\xi^* = 1\},\\
\mathfrak{L}_{\mathrm{res}} &= \{\xi : N_\xi > Y_X,\ 2 \le q_\xi^* \le Q_X^{\mathrm{cond}},\ \mathcal{R}(\xi)\},\\
\mathfrak{L}_{\mathrm{gen}} &= \{\xi : N_\xi > Y_X,\ 2 \le q_\xi^* \le Q_X^{\mathrm{cond}},\ \neg\mathcal{R}(\xi)\},\\
\mathfrak{H}_{\mathrm{res}} &= \{\xi : N_\xi > Y_X,\ q_\xi^* > Q_X^{\mathrm{cond}},\ \mathcal{R}(\xi)\},\\
\mathfrak{H}_{\mathrm{gen}} &= \{\xi : N_\xi > Y_X,\ q_\xi^* > Q_X^{\mathrm{cond}},\ \neg\mathcal{R}(\xi)\}.
\end{aligned}$$

Theorem 5.3 (Mutually exclusive complete sector partition)**.** *One has the disjoint union*

$$\boxed{\Xi_X = \mathfrak{N} \sqcup \mathfrak{S} \sqcup \mathfrak{C} \sqcup \mathfrak{L}_{\mathrm{res}} \sqcup \mathfrak{L}_{\mathrm{gen}} \sqcup \mathfrak{H}_{\mathrm{res}} \sqcup \mathfrak{H}_{\mathrm{gen}}.} \tag{33}$$

Consequently the literal resolved low window is the exact sum of the six potentially nonzero sector contributions. No exceptional, constant, resonant, or short fiber is hidden in the generic term.

Proof. Every key is first either structurally null or nonnull. A nonnull key is short or long. A long key has conductor 1, between 2 and Q_X^{cond}, or larger than Q_X^{cond}. In each nonconstant conductor range exactly one of $\mathcal{R}(\xi)$ and its negation holds. These successive binary and trinary alternatives are disjoint and exhaustive. □

5.3 The natural prime-modulus specialization

Corollary 5.4 (Five active sectors under the prime-modulus gap)**.** *Assume the hypotheses of theorem 4.4 and choose*

$$C_X \le Y_X, \qquad C_X \le Q_X^{\mathrm{cond}} < q_X.$$

Then:

(i) $q_\xi^* \le Q_X^{\mathrm{cond}}$ *is equivalent to* $q_X \mid \Omega_\xi$*;*

(ii) $q_\xi^* > Q_X^{\mathrm{cond}}$ *is equivalent to* $q_X \nmid \Omega_\xi$*;*

(iii) $\mathfrak{L}_{\mathrm{res}} = \varnothing$.

Thus the only active sectors are

$$\mathfrak{S}, \quad \mathfrak{C}, \quad \mathfrak{L}_{\mathrm{gen}}, \quad \mathfrak{H}_{\mathrm{res}}, \quad \mathfrak{H}_{\mathrm{gen}}.$$

Proof. The conductor gap gives (i) and (ii). If $\xi \in \mathfrak{L}_{\mathrm{res}}$, then

$$N_\xi \le q_\xi^* \le C_X \le Y_X$$

by lemma 4.10, contradicting the long condition $N_\xi > Y_X$. □

Corollary 5.5 (Four active sectors on canonical support)**.** *Under all hypotheses and threshold choices of corollary 5.4, use the literal common-target provenance proved in proposition 3.3. Then*

$$\mathfrak{L}_{\mathrm{res}} = \mathfrak{L}_{\mathrm{gen}} = \varnothing.$$

The complete structurally nonnull partition is

$$\boxed{\Xi_X \setminus \mathfrak{N} = \mathfrak{S} \sqcup \mathfrak{C} \sqcup \mathfrak{H}_{\mathrm{res}} \sqcup \mathfrak{H}_{\mathrm{gen}}.} \tag{34}$$

Every key in $\mathfrak{C}$ *has a phase constant for all low-window frequencies. Every key in* $\mathfrak{H}_{\mathrm{res}} \cup \mathfrak{H}_{\mathrm{gen}}$ *has conductor exactly* q_X*.*

Proof. Apply theorem 4.6: the only conductors are 1 and q_X. The first belongs to $\mathfrak{C}$ after short fibers are removed, and the second is above $Q_X^{\mathrm{cond}} < q_X$. □

Remark 5.6 (Why predicates must not be conflated)*.* Constancy implies resonance, but is separated first because it has no additive variation at all. Low conductor and resonance can overlap for arbitrary thresholds, which is why both $\mathfrak{L}_{\mathrm{res}}$ and $\mathfrak{L}_{\mathrm{gen}}$ appear in the general partition. Shortness depends on B and the native interval; it is not a frequency property. The hierarchical definition is what makes (33) genuinely disjoint.

6 Multiplier-mass accounting

We now state the accounting rule that turns the set partition into a valid physical return.

6.1 Normalized branch sums

Let

$$\beta = (\theta, c, \kappa)$$

be a soluble nonempty branch, with $b = c\kappa$, progression step B_β, and robustness content H_β. Combine the literal smooth weights and all native periodic masks with the two extraction masks into

$$\begin{aligned} \mathcal{W}_{\beta,X}(z) = \chi_\theta(\tau_\beta + B_\beta z) W_{\theta,X}(\tau_\beta + B_\beta z) \\ \times \mathbf{1}_{(H_\beta, \widetilde{D}_\beta(z))=1} \mathbf{1}_{(B_\beta, V_\beta(z))=1}. \end{aligned} \tag{35}$$

Define

$$S_{\beta,X}(r) = \sum_{z \in I_\beta^*} \mu(\widetilde{D}_\beta(z))\mu(V_\beta(z))\mathcal{W}_{\beta,X}(z)e(-\alpha_{\beta,r}z). \tag{36}$$

The exact outer coefficient, apart from the unit phase $\zeta_{\beta,r}$, is

$$A_{\beta,X} = c_{\theta,X}\mu(\kappa)\mu(B_\beta)\mu(H_\beta)\mathbf{1}_{c\kappa|P_\theta}. \tag{37}$$

On the canonical TPC-32 branch, $H_\beta = 1$.

For a sector $\mathfrak{E}$ in theorem 5.3, its exact contribution is

$$\mathcal{L}_X(\mathfrak{E}) = \sum_\beta A_{\beta,X} \sum_{\substack{r \in \mathscr{W}_X(R_X) \\ (\beta,r)\in\mathfrak{E}}} \mathfrak{m}_{K,X}(r)\zeta_{\beta,r}S_{\beta,X}(r). \tag{38}$$

The low window is the signed sum of these sector contributions.

6.2 The correct exceptional cost

Definition 6.1 (Branchwise multiplier mass)**.** For a branch β and a sector $\mathfrak{E}$, put

$$\mathfrak{M}_{\beta,X}(\mathfrak{E}) = \sum_{\substack{r \in \mathscr{W}_X(R_X) \\ (\beta,r)\in\mathfrak{E}}} |\mathfrak{m}_{K,X}(r)|. \tag{39}$$

Also put

$$\mathcal{A}_{\beta,X} = \sum_{z \in I_\beta^*} |\mathcal{W}_{\beta,X}(z)|\mu^2(\widetilde{D}_\beta(z))\mu^2(V_\beta(z)). \tag{40}$$

Theorem 6.2 (Multiplier-mass return inequality)**.** *For every sector $\mathfrak{E}$,*

$$\boxed{|\mathcal{L}_X(\mathfrak{E})| \le \sum_\beta |A_{\beta,X}|\,\mathcal{A}_{\beta,X}\,\mathfrak{M}_{\beta,X}(\mathfrak{E}).} \tag{41}$$

In particular, a frequency-cardinality bound is not a replacement for the multiplier mass in (41).

Proof. The unit phase has modulus one and

$$|S_{\beta,X}(r)| \le \mathcal{A}_{\beta,X}.$$

Apply the triangle inequality to (38), retaining the literal branch coefficient. The resulting inner sum is exactly (39). ☐

Remark 6.3 (Sufficient, not necessary). Theorem 6.2 is the correct absolute return of one exceptional sector. Cancellation may also occur between its frequencies or outer branches, so the inequality is not claimed to be necessary. What is invalid is to discard the sector using only the number of frequencies without proving a bound for the weighted sum that actually occurs.

6.3 Consequences of filter decay

TPC-86 supplies

$$|\mathfrak{m}_{K,X}(r)| \ll_K (1 + d_{q_X}(r,0))^{-K}.$$

Therefore

$$\mathfrak{M}_{\beta,X}(\mathfrak{E}) \ll_K \sum_{\substack{r \in \mathscr{W}_X(R_X) \\ (\beta,r)\in\mathfrak{E}}} (1 + d_{q_X}(r,0))^{-K}. \tag{42}$$

If every frequency in the sector has $d_{q_X}(r,0) \ge R_0$, then

$$\mathfrak{M}_{\beta,X}(\mathfrak{E}) \ll_K R_0^{1-K}. \tag{43}$$

There is also an exact description of the constant sector. When $q_X \mid \Omega_\beta$, write $\Omega_\beta = q_X \Omega'_\beta$ and put

$$\delta_\beta = \frac{c}{(c, \Omega'_\beta)}.$$

By (25), the constant frequencies are exactly the nonzero signed lifts $\widetilde{r}$ divisible by δ_β. Hence

$$\mathfrak{M}_{\beta,X}(\mathfrak{C}) \ll_K \delta_\beta^{-K}. \tag{44}$$

This can be small when δ_β grows, but can be of constant order when $\delta_\beta = 1$. It gives no cancellation inside the corresponding affine Möbius sum.

Corollary 6.4 (No filter-mass discount for a literal constant branch)**.** *On canonical common-target support,*

$$q_X \mid \Omega_\beta \quad\Longrightarrow\quad c \mid \Omega_\beta / q_X \quad\Longrightarrow\quad \delta_\beta = 1.$$

Hence, if the branch is long, every low-window frequency belongs to $\mathfrak{C}$ *and*

$$\mathfrak{M}_{\beta,X}(\mathfrak{C}) = \sum_{r \in \mathscr{W}_X(R_X)} |\mathfrak{m}_{K,X}(r)| \ll_K 1. \tag{45}$$

The bound is uniform but gives no decaying factor.

Proof. Canonical support gives $c \mid \Omega_\beta$. Since $(c, q_X) = 1$, the additional divisibility $q_X \mid \Omega_\beta$ implies $c \mid \Omega_\beta / q_X$. Thus $\delta_\beta = 1$. Theorem 4.6 makes the phase constant for every r, and the filter decay is summable for $K \geq 2$. $\square$

Remark 6.5 (One heavy frequency can dominate)*.* The decay in (42) distinguishes a singleton near frequency 1 from many far frequencies. Their cardinalities point in the opposite direction from their possible costs. This is why the exceptional ledger must use $\mathfrak{M}_{\beta,X}$.

6.4 What each sector asks for

Sector	Exact structure	Missing input or valid return
$\mathfrak{S}$	$N_\xi \leq Y_X$, often because B_ξ is large	Return full branch mass times multiplier mass; prove it fits the physical budget.
$\mathfrak{C}$	$cq_X \mid \widetilde{r}\Omega_\xi$	No phase cancellation; use Möbius correlation, outer coherence, or the exact mass bound (44).
$\mathfrak{L}_{\mathrm{res}}, \mathfrak{L}_{\mathrm{gen}}$	Exact period $q_\xi^* \leq Q_X^{\mathrm{cond}}$	Split with all masks retained; periodicity alone does not control the Möbius product. These sectors are empty on literal canonical support.
$\mathfrak{H}_{\mathrm{res}}$	$N_\xi \Vert\alpha_\xi\Vert \leq 1$	Near-constant phase; requires an arithmetic or outer-coherence estimate.
$\mathfrak{H}_{\mathrm{gen}}$	High conductor and $N_\xi \Vert\alpha_\xi\Vert > 1$	Natural candidate for hybrid additive/Möbius cancellation, still an L2 theorem.

Table 1: Exact sector meaning and the remaining proof obligations.

7 Physical interface and audited claim boundary

7.1 The exact advance

Theorem 7.1 (Audited content-progression and resonance ledger)**.** *Import the established TPC-93 Theorem 3.3, Corollary 3.4, and Theorem 3.5 (Complete Decorated Affine Export), including their lossless inverse provenance and multiplicity reconstruction. Then, for every literal TPC-93 content-resolved low-window term:*

(i) the raw reparametrized determinant is Bh_0;

(ii) the forced factor $U(\tau + Bz) = BV(z)$ restores determinant h_0;

(iii) in an arbitrary extension, the only further primitive determinant is h_0/H, where $H = (B, h_0) \mid h_0$;

(iv) on canonical TPC-32 support, b is an actual common target divisor, so $(b, h_0) = 1$, $H = 1$, and the primitive determinant remains the prescribed h_0;

(v) squareful B branches vanish, while squarefree branches retain the exact sign and coprimality mask in (13);

(vi) the additive phase retains oriented slope $\varepsilon_\theta \Omega = \varepsilon_\theta \ell v \sigma B$ and has exact conductor (21);

(vii) canonical target provenance gives $b \mid \Omega$, so the literal conductor is exactly 1 or q_X, with no nonconstant low-conductor sector;

(viii) the prime-modulus conductor gap and the disjoint partitions (33) and (34) hold; and

(ix) every absolute exceptional return is charged by (41), not by frequency cardinality.

Proof. Items (i)–(v) are proposition 2.1, theorems 2.2, 2.5 and 3.1, and corollary 3.2. Items (vi)–(ix) are proposition 3.3, theorems 4.4, 4.6, 5.3 and 6.2, and corollary 5.5. □

7.2 Promotion requirements

The exact decomposition becomes a zero-mode estimate only after the complete signed sum satisfies the TPC-92 target

$$|Z_X| \le X^{-\eta_Z + o(1)} Q_X^2$$

with all short, constant, low-conductor, resonant, generic, content, and tail returns included [4]. Sufficient sectorwise proofs may be combined, but cancellation across sectors is also allowed if established for the exact coherent outer sum.

Any future theorem must retain:

(1) the signed multiplier $\mathfrak{m}_{K,X}(r)$;

(2) the polarization orientation ε_θ and the signed lift $\widetilde{r}$ in every modulus-cq_X character;

(3) the exact (c, κ) distinction and the phase modulus cq_X, not bq_X;

(4) the factor $\mu(B)$ and mask $\mathbf{1}_{(B,V)=1}$;

(5) the original and reduced determinant labels;

(6) both polarizations, every native interval and source key, and one global normalization; and

(7) every returned sector under multiplier-mass accounting.

7.3 Separate ledgers

No exponent is gained in this paper. The literal L1 identification above imports the cited TPC-93 lossless-provenance theorem and its explicit inverse ledger. If later work proves a zero-mode saving η_Z and a determinant-mass loss λ_D, the compatibility condition remains

$$\lambda_D \le 2\eta_Z.$$

Independently, all physical localization, short-fiber, completion, and reassembly losses must satisfy the strict endpoint rule

$$\Lambda_{\text{phys}} < \frac{1}{400}.$$

The same sector loss cannot be entered in both ledgers. Equality in the physical ledger is not a reserve.

7.4 Finite regression certificate

The deterministic script `experiments/tpc94_certificate.py` performs 2,275,219 exact integer checks:

normalization and Möbius extraction	205,885,
two-polarization phase splitting	1,989,680,
signed lifts and conductors	57,954,
progression lengths	21,600,
sector boundaries	100.

It also records the abstract representative-ambiguity witness

$$(q, c, \Omega) = (3, 2, 1),$$

for which the representatives 2 and -1 would give conductors 3 and 6 without (16). These checks are reproducible finite identity regressions. They are not evidence for asymptotic cancellation and do not independently re-run the upstream TPC-93 source–child inverse certificate.

7.5 What is not proved

- No constant, resonant, low-conductor, or short sector is shown to have the required aggregate saving.
- No fixed-form theorem is promoted to the literal growing family.
- No oscillatory estimate for the high-conductor generic sector is proved.
- No determinant-energy lower bound or full-block reassembly is supplied.
- No fixed-h_0 prime-pair estimate, parity breach, or twin-prime conclusion follows.

Remark 7.2 (Next theorem targets)*.* The cleanest next attempts are now explicit: prove that the literal constant and low-conductor multiplier masses are affordable; return all B-shortened fibers under the global normalization; and seek a uniform weighted two-form estimate on the high-conductor generic sector. A rigorous lower bound showing that one returned sector already exceeds its allowable budget is an equally valid negative certificate for that route.

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